

FTR

Core Formalization

Future trajectory representation for autonomous-agent runtimes:
possible futures, disagreement, future information requirements, proof obligations, risk, resolution and calibration.

$$\mathcal{F}_t = (P_t(\tau), D_t, \mathcal{H}_t, \Omega_t^*, P_t^{\text{proof}}, R_t, \Gamma_t)$$

Status: retired by its own criteria, 19 September 2026. Every FTR sub-claim either failed the kill threshold written for it, reduced to prior art, or proved not evaluable by its one designed experiment. FTR was never a validated theory and is no longer a research direction. This document is kept as the record of what was formalized and why it died. The one question that outlives it — which facts must be kept before an action because a later proof will need them (Ω^* , *preserve*) — is marked **OPEN** and no longer carries FTR's name.

Version	2026-09-19 · supersedes the undated working specification
Evidence base	MØBIUS repository at <code>61f7100</code> (R3 handoff), evidence directories <code>r1/f9b27dd</code> , <code>r2/f9b27dd</code> , <code>p4-expectations/ba4f291</code> ; research-lead records of 18–19 Sep 2026
Companion post	marcyy.me/posts/mobius-a-correct-world-is-not-a-proven-one
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0 • What changed in this version

- **FTR retired itself.** Reopened on 19 September as a looser question, then put through its own tests: no task class where planning and replanning fall short; every component reduced to a known object; the only designed experiment not evaluable. Retired the same day. By FTR's own logic — find early which beliefs about the future will not survive — this is its success, not a failure (§15).
- **Evidence horizon, corrected.** The previous version implied information that becomes *unrecoverable* after an action. In every R3 P3 journal with a transition-shaped criterion, the pre-state *was* captured before the first effect; it was absent from the judge's *view*, not from the journal. The unrecoverable variant has not been observed and is kept as a hypothesis (§6).
- **Transition witness, downgraded.** Retired as a new primitive (before-image + provenance edge + past-time operator). Kept as a candidate *representation* (§7).
- **Measured additions.** Hand-coded cause of all 74 `cannot-tell` verdicts; a manual audit of all 20 `established` verdicts (one unsupported); certificate closure measured at 0 of 11 (§12).
- **Vacuity made explicit.** R3 P3's "0 false completions" had zero opportunities to fail (39/39 rows passed their checker) and is recorded as not evaluable (§15).
- **Write-set anticipation.** Stated precisely: a kill threshold fixed before the computation, on an experiment-arm corpus, not a workload (§15).

1 • Motivation — what the evidence points at

MØBIUS's completion plane was the place its questions moved in September 2026. A separate model call (the *judge*, ADR 0043) decides whether a criterion no assertion checks is `established`, `not-established` or `cannot-tell`, from the observations it is shown (ADR 0045: the binding's evidence plus the run's latest non-stale observation of each other resource). In R3 P3 (`claude-opus-5` , three repetitions, 39 valid rows) the judge ruled 114 times: 20 / 20 / 74.

what a <code>cannot-tell</code> reason says is missing	n
prior state only	40
prior state and an unread current resource	5
an unread current resource only	28
judge call timed out	1

The 45 reasons that cite a missing past sit in three unrelated task shapes — rename (10/10), edit-to-spec with "nothing else changed" (20/22), split a document (15/27) — and in none of the other three tasks (0/15). Their criteria are predicates over a *transition*, not a state:

$$C \neq f(W_{\text{post}}), \quad C = f(W_{\text{pre}}, A, W_{\text{post}})$$

FTR's surviving question follows from this: **can a runtime know, before it acts, what a future decision or a future proof will need it to have kept?**

2 • Runtime state and history

$$X_t = (W_t, H_t, G_t, M_t, C_t, E_t, \Gamma_t)$$
$$H_t = (O_0, A_0, E_1, O_1, A_1, E_2, \dots, A_{t-1}, E_t, O_t)$$

History is not state. A current state is a function of history, and not an injective one:

$$H_t \not\equiv W_t, \quad W_t(h_i) = W_t(h_j) \not\Rightarrow h_i = h_j$$

This matters whenever a criterion concerns preservation, movement, non-modification, provenance or causality — any past-to-present relation.

3 • Possible futures as the object

$$\tau_{t:t+k} = (S_t, A_t, E_{t+1}, O_{t+1}, S_{t+1}, \dots, A_{t+k-1}, E_{t+k}, O_{t+k}, S_{t+k})$$
$$P_t(\tau) = P_\theta(\tau_{t:t+k} \mid H_t, W_t, G_t, C_t, M_t), \quad \mathcal{T}_t = \{(\tau_i, p_i)\}_{i=1}^N, \quad p_i \geq 0, \quad \sum_{i=1}^N p_i = 1$$

The target is a distribution over structured trajectories — actions, effects, observations, failures, retries, human intervention, evidence capture, verification, completion — and never "the next action".

4 • FTR is not a plan

$$\mathcal{F}_t \neq \text{Plan}_t, \quad P(\tau \mid X_t, \text{Plan}_t) \neq \mathbf{1}[\tau = \text{Plan}_t]$$

A plan is prescriptive: *what do I intend to do?* FTR is anticipatory: *which futures remain reachable from here, where do they diverge, what will they require, and what should the runtime preserve or inspect now because of that?* It must admit futures in which the plan fails, branches, stalls on a person, loses its evidence or is replaced.

Kill condition: an FTR implementation that reduces to the planner under another name.

5 • Future disagreement and uncertainty

Let $z \in Z$ be equivalence classes of materially distinct outcomes or trajectory shapes.

$$D_t = 1 - \sum_{z \in Z} P(z \mid X_t)^2, \quad \mathcal{H}_t = - \sum_{i=1}^N p_i \log p_i$$

D_t is the Gini–Simpson index over outcome classes: $D_t \approx 0$ when rollouts agree, $D_t \rightarrow 1$ when they scatter. It does not require knowing which future is correct. PRIOR ART Sample disagreement as an uncertainty estimate: self-consistency (Wang et al., ICLR 2023), semantic entropy over equivalence classes (Kuhn, Gal, Farquhar, ICLR 2023), conformal ask-for-help sets (Ren et al., CoRL 2023). OPEN Whether divergence is a useful *runtime trigger* — for observation, preservation, a question to a person, or delaying an irreversible effect.

6 • Future information requirements and proof obligations

Let $E_\Omega(h)$ be the evidence history h exposes when the fact set Ω is captured, and $C(h) \in \{0, 1\}$ whether the criterion holds in h . The cheapest sufficient capture:

$$\Omega_t^* = \arg \min_{\Omega} \text{Cost}(\Omega) \quad \text{s.t.} \quad \forall h_i, h_j : E_{\Omega}(h_i) = E_{\Omega}(h_j) \Rightarrow C(h_i) = C(h_j)$$

Equivalently $C(h_i) \neq C(h_j) \Rightarrow E_{\Omega}(h_i) \neq E_{\Omega}(h_j)$: no two possible histories may look identical under the evidence while disagreeing on the criterion. This is the indistinguishability test for evidential soundness. For the rename, $\Omega_t^* = \{\text{digest}(\text{notes}_{\text{pre}})\}$.

$$P_t^{\text{proof}}(a) = P(\neg \text{Provable}(C_{t+k}) \mid a, X_t) \quad \text{is not} \quad P(\text{TaskFailure} \mid a)$$

$$\Delta_{\text{preserve}} = P_t^{\text{proof}}(a) - P_t^{\text{proof}}(a \mid \text{preserve}(\Omega)), \quad \text{preserve}(\Omega) \text{ worthwhile iff } \Delta_{\text{preserve}} \cdot V_{\text{proof}} > \text{Cost}(\Omega)$$

An action may be operationally safe and epistemically dangerous: $P(\text{succeeds} \mid a) \approx 1$ while $P(\text{Provable}(C) \mid a) \ll 1$. MEASURED T4 is one real instance of the second quantity collapsing *for the judge's view* after a move, although the journal still held the proof. OPEN Whether Ω^* or P^{proof} can be estimated before an action, for criteria written in natural language. OPEN The *unrecoverable* case — a fact that exists only before the action — has a minimal construction (move with no prior read) and no observed instance.

7 • Transition witnesses

$$\omega_t = (\phi_{\text{pre}}(W_t), a_t, \phi_{\text{post}}(W_{t+1}), \rho(W_t, a_t, W_{t+1})), \quad \mathcal{T}_{t \rightarrow t+1} = (W_t, A_t, W_{t+1}, R_t)$$

$$\rho_{\text{move}} : \text{digest}(\text{src}@t) = \text{digest}(\text{dst}@t+1) \wedge \text{exists}(\text{src}@t+1) = 0 \wedge \text{exists}(\text{dst}@t+1) = 1 \Rightarrow \omega_t \models C$$

RETIRED as a new primitive: it is a before-image, a provenance edge and a past-time temporal operator (Havelund & Roşu, TACAS 2002; Snodgrass & Ahn 1986; Buneman et al. 2001; Green et al. 2007). CANDIDATE as a representation. OPEN The systems question is narrower: which transitions an agent runtime must make first-class evidence, and how it decides that *before* acting. A digest shows equal bytes, not the causal path; for "moved, not rewritten" it is insufficient.

8 • Risk over possible futures

$$R_t(a) = \underbrace{\mathbb{E}_{\tau \sim P_t}[L(\tau, a)]}_{\text{bad outcome}} + \underbrace{\lambda_D D_t(a)}_{\text{divergence}} + \underbrace{\lambda_P P_t^{\text{proof}}(a)}_{\text{unprovable completion}}$$

9 • Runtime metareasoning

$$\mathcal{X}_t = \{\text{act}, \text{observe}, \text{test}, \text{ask}, \text{fork}, \text{replan}, \text{preserve}, \text{wait}\}$$

$$V(\mathcal{F}_t) = -[\alpha \mathcal{H}_t + \beta D_t + \gamma R_t + \delta P_t^{\text{proof}}]$$

$$x_t^* = \arg \max_{x \in \mathcal{X}_t} [\mathbb{E}[V(\mathcal{F}_{t+1}) \mid x, X_t] - V(\mathcal{F}_t) - \lambda_C \text{Cost}(x) - \lambda_L \text{Latency}(x)]$$

PRIOR ART Value of information (Howard 1966) and rational metareasoning (Russell & Wefald 1991) supply the decision rule; speculative execution (Nightingale et al., SOSp 2005) is the systems precedent for acting on a predicted future. The only candidate contribution is the *state* the rule is computed over — in particular *preserve*, choosing what to keep because of what a future proof will need. Its cheapest rival is a wholesale snapshot, which on a filesystem costs almost nothing.

10 • Forecast identity, resolution and calibration

$$f_i = (W_i, B_i, \tau_i, p_i, \Omega_i, T_i), \quad \text{Valid}(f_i, t) \iff B_i \simeq B_t \wedge W_i \simeq W_t$$

W_i	world version the forecast was made against
B_i	basis: observations, evidence, assumptions, runtime state
τ_i	forecast trajectory or outcome class
p_i	probability / confidence
Ω_i	predicted future information requirement
T_i	resolution condition / horizon

$$\text{Resolve}(f_i, O_{i+1:t}) \in \{\text{satisfied}, \text{violated}, \text{void}\}$$

satisfied — the basis held and the predicted event occurred. **violated** — the basis held and it did not. **void** — the world or premise changed so the forecast no longer applies; not a prediction error. The same version-binding appears in MØBIUS's shipped approval revalidation (`reusable` | `superseded` | `unevaluable`) and in published commit-time authorization (CommitGuard, arXiv 2607.10487) — it is `PRIOR ART` as a pattern.

$$y_i = \mathbf{1}[\text{Resolve}(f_i) = \text{satisfied}], \quad \text{BS} = \frac{\sum_{i: r_i \neq \text{void}} (p_i - y_i)^2}{|\{i : r_i \neq \text{void}\}|}$$

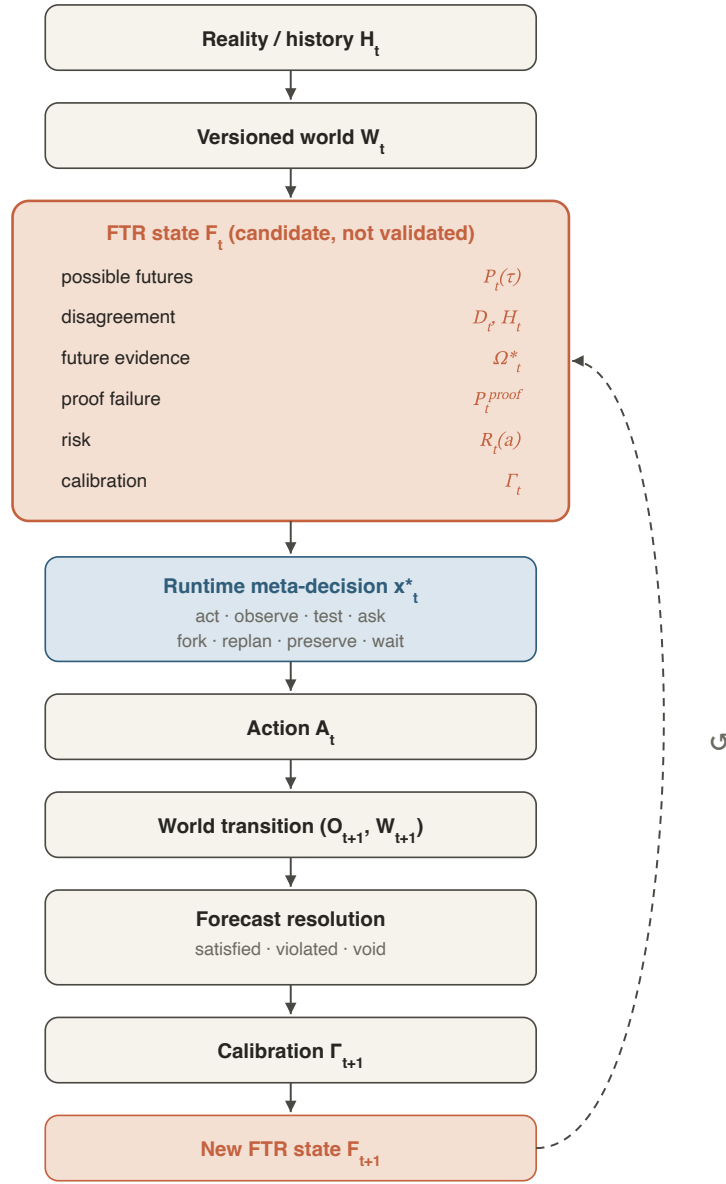
$$\Gamma_t = \text{CalibrationHistory}(f_1, \dots, f_t), \quad \text{Trust}_{t+1} = g(\Gamma_t)$$

`NOT EVALUABLE` on existing data. R3 P4 resolved 1,345 model-written `expected` assertions from 104 journals: they are bare Booleans. A constant forecast has one bin, so the Murphy resolution term is zero by definition whatever the outcomes, and a kill criterion "resolution ≈ 0 " fires on arithmetic. Calibration is a *requirement* on any future FTR experiment, not a result.

11 • The candidate object and its loop

$$\mathcal{F}_t = P_\theta(\tau_{t:t+k}, \Omega_{t:t+k}, R_{t:t+k} \mid H_t, W_t, G_t, C_t, M_t)$$

$$X_t \rightarrow \mathcal{F}_t \rightarrow x_t^* \rightarrow A_t \rightarrow (O_{t+1}, W_{t+1}) \rightarrow \text{Resolve} \rightarrow \Gamma_{t+1} \rightarrow \mathcal{F}_{t+1}$$



Only the Boolean expectation layer exists in code today; nothing here is implemented as FTR.

Figure 1. FTR as a candidate closed loop. Only a Boolean expectation layer exists in code.

12 · Completion soundness and certificate closure

Let J_i be the evidence the judge was actually shown for criterion i and C_i the evidence the final completion record keeps.

closure: $J_i \subseteq C_i$

replay: $\text{Judge}(\text{Criterion}_i, C_i) = \text{OriginalDecision}_i$

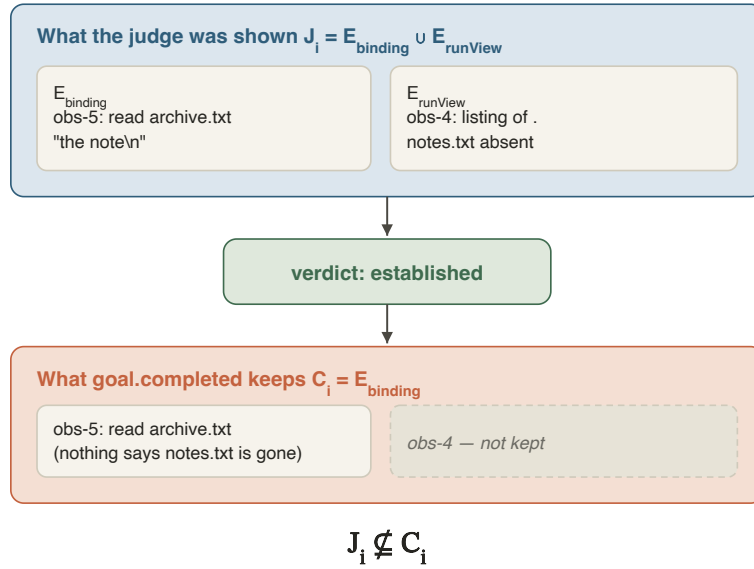
soundness: $\text{Judge}(C, E) = \text{established} \Rightarrow E \models C$

world correctness $\not\equiv$ evidential correctness

MEASURED In MØBIUS at `f9b27dd` the judge is shown $J_i = E_{\text{binding}} \cup E_{\text{runView}}$, while `#bindClaimed` records $C_i = E_{\text{binding}}$ and `goal.completed` carries it. Across R3 P3, 11 judge-bound criteria appear in completed runs;

$J_i \subseteq C_i$ holds in **0**; in 8 the omitted evidence carries part of what the judge's own reason cites. The journal's `criterion.judged` event still names J_i : the gap is in the certificate, not the log. Replay has not been run.

MEASURED Manual audit of all 20 `established` verdicts: 19 supported by what the judge was shown (3 only on a literal reading); **1 not** — R1 T4 repetition 3, where the judge answered `cannot-tell` correctly, was then shown only a read of `archive.txt`, and answered `established` with a reason that never mentions the former content. One in twenty is an existence proof, not a rate.



R3 P3: 11 judge-bound criteria in completed runs; $J \subseteq C$ in 0 of 11.
 In 8 of the 11 the omitted evidence carries part of what the judge's own reason cites.
 The journal's `criterion.judged` event still names J — the gap is in the certificate, not in the log.

Figure 2. The judge's basis against the completion record, T4 repetition 3.

13 · A worked temporal example

Criterion C : *archive.txt exists with the former content of notes.txt, and notes.txt no longer exists.*

$$E_{\text{post}} = \{ \text{archive}_{t+1} = \text{"the note"}, \neg \text{exists}(\text{notes}_{t+1}) \}$$

$$h_1 : \text{notes}_t = \text{"the note"} \rightarrow \text{archive}_{t+1} = \text{"the note"} \quad C(h_1) = 1$$

$$h_2 : \text{notes}_t = \text{"other"} \rightarrow \text{archive}_{t+1} = \text{"the note"} \quad C(h_2) = 0$$

$$E_{\text{post}}(h_1) = E_{\text{post}}(h_2) \Rightarrow E_{\text{post}} \not\models C$$

$$\Omega = \{ \text{digest}(\text{notes}_t) \} : \text{digest}(\text{notes}_t) = \text{digest}(\text{archive}_{t+1}) \wedge \neg \text{exists}(\text{notes}_{t+1}) \Rightarrow C$$

In the real journal (r1/f9b27dd/P3-control-b/T4-rename/rep-3) both digests are `sha256:6862e454af825d8f...`: event 7, before the move, and event 17, after it.

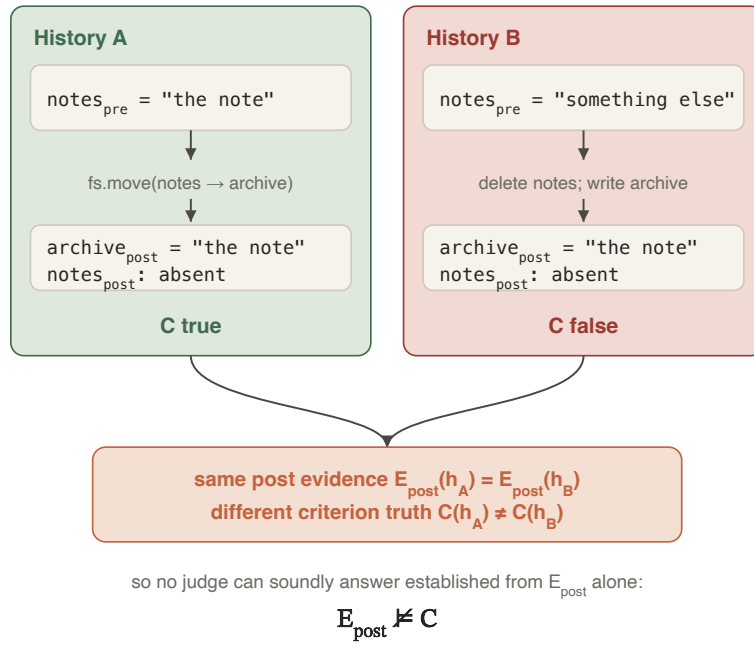


Figure 3. Same post-state evidence, opposite criterion truth.

14 • Notation

W_t	versioned external world state at t
H_t	execution history: observations, actions, effects
G_t, C_t	active goals; completion criteria
M_t	agent / runtime internal state: plan, tasks, subagents, tool context
E_t	evidence currently available (to whom — journal, proposer, judge — must be stated)
Γ_t	forecast resolution and calibration history
$\tau, P_t(\tau), \mathcal{T}_t$	trajectory; distribution over trajectories; sampled approximation
$D_t, \mathcal{H}_t$	disagreement (Gini–Simpson over outcome classes); entropy
Ω_t^*	cheapest fact set that keeps criterion-distinct histories distinguishable
P_t^{proof}	probability that completion becomes unprovable
$R_t(a)$	risk of action a over futures
$\mathcal{X}_t, x_t^*$	meta-action set; chosen meta-action
$f_i, \text{Resolve}, \text{BS}$	forecast instance; resolution; Brier score over non-void forecasts
J_i, C_i	evidence the judge was shown; evidence the completion record keeps

15 · Research status, 19 September 2026

claim / component	status	basis
FTR overall	RETIRED BY ITS OWN CRITERIA	reopened as a question, then retired on 19 Sep: every sub-claim failed its threshold, reduced to prior art, or could not be evaluated
future prediction itself as novelty	RETIRED	predictive runtime verification (Lindemann et al., arXiv 2211.01539)
exact prediction of future actions	UNSUPPORTED	not the surviving claim
read-only pre-write write-set anticipation	KILLED	median read coverage 0.25 (33 of 69 zero) against a recall < 0.5 threshold fixed before computing; experiment-arm corpus, so only the shape split (edit vs create/rename) generalises
Boolean expected → Brier / resolution	NOT EVALUABLE	constant forecasts; resolution zero by definition
rollout disagreement as runtime signal	OPEN	as uncertainty: prior art; as trigger: untested
future information requirement Ω^*	OPEN	outlives FTR under its own name; no estimator yet
anticipatory evidence preservation	OPEN	experiment E4
proof-failure prediction	OPEN	one instance of the quantity; no predictor
transition witness	CANDIDATE	retired as a primitive; kept as representation
evidence horizon (past out of judge's view)	MEASURED OPEN	45 / 74 cannot-tell; three task shapes; general version open
unsupported establishment	MEASURED OPEN	1 clear case in 20; rate unknown (E1)
certificate closure / replayability	MEASURED OPEN	0 / 11 closed; replay not run (E3)
version-bound decisions and forecasts	PRIOR ART	implemented and measured in MØBIUS; published (CommitGuard, PlanFence)
value of information / computation	PRIOR ART	decision machinery, not FTR novelty
three-valued outcomes / partial observation	PRIOR ART	not a novelty claim
"0 false completions" (R3 P3)	NOT EVALUABLE	0 opportunities: 39 / 39 valid rows passed their checker

16 • Prior-art boundary

Before any FTR component is claimed as new it must be separated from: predictive runtime verification and conformal prediction for STL (arXiv 2211.01539); past-time LTL monitoring (TACAS 2002) and runtime verification generally (Leucker & Schallhart, JLAP 2009); temporal databases (Snodgrass & Ahn 1986); provenance (ICDT 2001; PODS 2007); proof-carrying code (POPL 1997); ATMS (de Kleer 1986); value of information (Howard 1966) and rational metareasoning (Russell & Wefald 1991); speculative execution (SOSP 2005); sampling-based uncertainty (arXiv 2203.11171, 2302.09664, 2307.01928); commit-time and dependency-scoped validation for LLM agents (arXiv 2607.10487, 2609.03340) and read-set reconstruction (arXiv 2605.17076); false success in LLM agents (arXiv 2606.09863).

17 • Candidate invariants

1. A forecast carries explicit W, B, T, p .
 2. Invalidation of a forecast's basis resolves it as *void*, never as an ordinary error.
 3. Before a world-changing action, evidence-loss and proof-loss are evaluated.
 4. Preserved evidence is attributable to a pre-action world version.
 5. **established** $\Rightarrow E \models C$ for the evidence actually shown.
 6. The completion record contains the decision basis: $J_i \subseteq C_i$.
 7. A forecast's influence on action is weighted by measured calibration, never by its rhetoric.
 8. Every existence claim is reported with its opportunity count; zero opportunities is "not tested".
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18 • Open experiments

E1 • Evidential soundness replay. Freeze criterion, prompt, evidence and model for T4 post-only evidence ($E \not\models C$); re-ask N times.

$$\hat{p}_{\text{est}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{Judge}_i(C, E) = \text{established}]$$

$\hat{p}_{\text{est}} > 0$ with $E \not\models C$: unsupported establishment is a property of the judge under that evidence. $\hat{p}_{\text{est}} = 0$: repetition 3 is not reproducible from its inputs.

E2 • Evidence-horizon intervention.

$$A_0 = E_{\text{post}}, \quad A_1 = E_{\text{post}} \cup E_{\text{pre}}, \quad A_2 = E_{\text{post}} \cup \omega_{\text{transition}}, \quad A_3 = E_{\text{post}} \cup \text{proposer self-claim}$$

Measure $P(\text{established} \mid A_k)$, $P(\text{cannot-tell} \mid A_k)$, $P(\text{unsupported} \mid A_k)$. Hypothesis: $A_1, A_2 \gg A_0$ for justified establishment and $A_3 \approx A_0$. If A_3 moves the judge like A_1 , the judge reads assertions rather than evidence.

E3 • Certificate closure audit and replay. $\chi_i = \mathbf{1}[J_i \subseteq C_i]$, ClosureRate = $\frac{1}{N} \sum_i \chi_i$ (0/11 today); re-judge from C_i alone before any fix, so the fix has a baseline.

E4 • Anticipatory evidence preservation. Before a world-changing action, offer *preserve* or *proceed*; trigger $\mathbf{1}[P_t^{\text{proof}}(a) > \eta]$. Compare with preserve-nothing and preserve-everything (snapshot) baselines on tasks where a later criterion depends on pre-state and matched tasks where it does not:

$$\Delta CT = CT_{\text{base}} - CT_{\text{FTR}}, \quad \Delta UE = UE_{\text{base}} - UE_{\text{FTR}}, \quad \Delta \text{Cost} = \text{Cost}_{\text{FTR}} - \text{Cost}_{\text{base}}$$

Kill: no reduction in cannot-tell or unsupported establishment beyond the snapshot baseline at comparable cost, or no generalisation beyond the three task shapes. Every arm reports its opportunity count.

19 • Evidence index

R2 pilot (0/6 checker, 5/6 declared complete)	<code>evidence/r2/de5b79b</code>
ADR 0043 re-run (false 0/6, runtime 0/6; R1 6/7)	<code>evidence/{r2,r1}/33e599e</code>
R3 P3 (39 valid rows, 114 judgements)	<code>evidence/r2/f9b27dd/{P3-judge-run,P3-judge-run-b}</code> , <code>evidence/r1/f9b27dd/{P3-control,P3-control-b}</code>
R3 P4 (1,345 assertions, 104 journals)	<code>evidence/p4-expectations/ba4f291</code>
layered determinism (0.923 / 0.795 / CV 0.129)	re-derived from R3 P3 journals, 13 tasks × 3
judge and certificate code	<code>packages/runtime-core/src/loop.ts</code> — <code>#bindClaimed</code> , <code>#established</code> , <code>#runView</code>

*What is clearer now is not what the answer is,
but what it is not — and which questions are worth the next experiment.*

Marc Ellington

MØBIUS Research

A handwritten signature in black ink, reading "Marc Ellington". The script is fluid and cursive, with the first letters of "Marc" and "Ellington" being capitalized and prominent.

19 September 2026